MATH-18A College Algebra **Exam 2 Practice**

The exam will have around 7 questions, and one optional bonus question on the last page. You are allowed a non-graphing calculator and ONE 3×5 notecard. **Show all your work** and **box your answers**.

This answer is to check numbers only; these are **NOT full credit answers**. You must follow all directions (show work, etc) to receive full credit.

Section 2.1 (4 points) Multiply and combine like terms.

1. $f(x) = (3x+2) \cdot (2x^3 - 2x^2 + 4) = 6x^4 - 2x^3 - 4x^2 + 12x + 8$

2. $g(x) = (x-3) \cdot (x^2 + 3x + 4) = x^3 - 5x - 12$

Section 2.2

3. (6 points) Find **all real solutions** of $15x^2 + 7x - 2 = 0$. Your final answer should include the **factored form** of the equation AND the x-intercepts (real zeroes).

$15x^2 + 7x - 2 = (5x-1)(3x+2)$

$x = -2, 1/5$

For more practice, repeat with the following:

4. $2x^2 - 5x - 3 = 0$

$12x^2 - 5x - 3 = (2x+1)(x-3)$

$x = 3, -1/2$

5. $4x^2 - 8x + 53 = 0$

$$4x^2 - 8x + 53 = \left(x - \frac{8 - \sqrt{-784}}{8}\right)\left(x - \frac{8 + \sqrt{-784}}{8}\right)$$

$x = \dfrac{8 \pm \sqrt{-784}}{8}$ so there are no real solutions

6. $3x^2 - 5x - 3 = 0$

$$3x^2 - 5x - 3 = \left(x - \frac{5 + \sqrt{61}}{6}\right)\left(x - \frac{5 - \sqrt{61}}{6}\right)$$

$x = \dfrac{5 \pm \sqrt{61}}{6}$

Section 2.3

7. (6 points) Find **all real solutions** to $\frac{x}{x+1} - 5 = \frac{3}{x}$. State which values of x are not allowed, including any extraneous solutions.

$x = -\frac{1}{3}, -\frac{3}{2}$ and $x \neq 0, -1$

More more practice, repeat with the following:

8. $\frac{1}{3} + \frac{1}{x-3} = \frac{3}{x(x-3)}$

$x = -3$ and $x \neq 0, 3$

9. $\frac{3}{x-3} + \frac{x}{x-3} = \frac{3}{2}$

$x = 15$ and $x \neq 3$

10. $\frac{3}{x+1} + \frac{2}{x-4} = \frac{4x-11}{(x+1)(x-4)}$

Since $x \neq 4, -1$, there is no solution

Section 2.4

11. (6 points) Write the quadratic function $f(x) = 9 + 6x - 3x^2+$ in **vertex form**, and write the vertex as a **coordinate pair**.

$f(x) = -3(x-1)^2 + 12$, vertex is $(1, 12)$

More more practice, repeat with the following:

12. $f(x) = -2x^2 + 12x + 24$

$f(x) = -2(x-3)^2 - 6$, vertex at $(3, -6)$

13. $f(x) = 2x^2 + 6x + 1$

$f(x) = 2\left(x + \frac{3}{2}\right)^2 - \frac{7}{2}$, vertex as $\left(-\frac{3}{2}, -\frac{7}{2}\right)$

Section 2.5

14. (6 points) A ball is launched from atop a 101-foot cliff with an initial velocity of 116 ft/s. Round to the nearest tenth of a second. Make sure to include units in your answers. (Note: if a is not given in the problem, use -16 ft/s^2 or -4.5 m/s^2 depending on the units in the problem.)

a) What is the **height** above the ground when the object is launched? 101 ft.

b) **When** does the object hit the ground after launch? 8.0 s

c) What is the **maximum height** of the object? 311.3 ft

d) **When** does the object reach its maximum height? 3.6 s

More more practice, repeat with the following:

15. A water balloon is catapulted into the air so that its height h, in meters, after t seconds is $h(t) = -5t^2 + 30t + 10$.

a) 10 ft

b) 6.3 s

c) 55 ft

d) 3 s

16. The height h in feet of a projectile launched vertically upward from the top of a 96-foot-tall tower when time $t = 0$ is given by $h(t) = 96 + 80t - 16t^2$.

a) 96 ft

b) 6.0 s

c) 196 ft

d) 2.5 s

Section 2.6

17. (6 points) Divide $3x^3 - 7x^2 - 18x - 8$ by $3x + 2$ using long division. Express your final answer in **division algorithm format**:

"dividend = (divisor)(quotient) + remainder"

$3x^3 - 7x^2 - 18x - 8 = (3x + 2)(x^2 - 3x - 4)$ with remainder zero

More more practice, repeat with the following:

18. $x^4 + 5x + 2$ by $x - 2$

$x^4 + 5x + 2 = (x - 2)(x^3 + 2x^2 + 4x + 13) + 28$

19. $3x^3 - 7x^2 - 18x - 8$ by $x + 1$

$3x^3 - 7x^2 - 18x - 8 = (x + 1)(3x^2 - 10x - 8)$ with remainder zero

Section 2.7

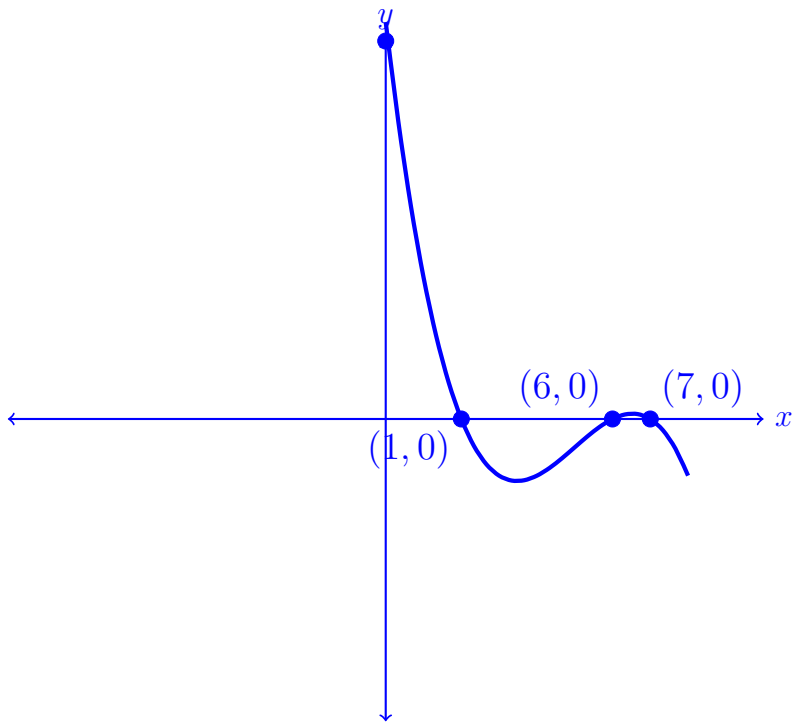
20. (6 points) Consider the polynomial $y = (-2x + 12)(x - 1)(x - 7)$.

a) What are the **real zeroes** and their **multiplicites**?
$(0, 1), (0, 6), (0, 7)$ all with multiplicity 1

b) Is the y-intercept **above or below** the x-axis (is it positive or negative)?
above

c) What is the **end behavior**?
leading term: $(-2x)(x)(x) = -2x^3$
negative, so RHS goes down
degree odd, so LHS is opposite (goes up)

d) **Sketch** the graph on the axes below. You do not need to draw it to scale, but clearly label the real zeroes, and clearly show the end behavior of the graph.



More more practice, repeat with:

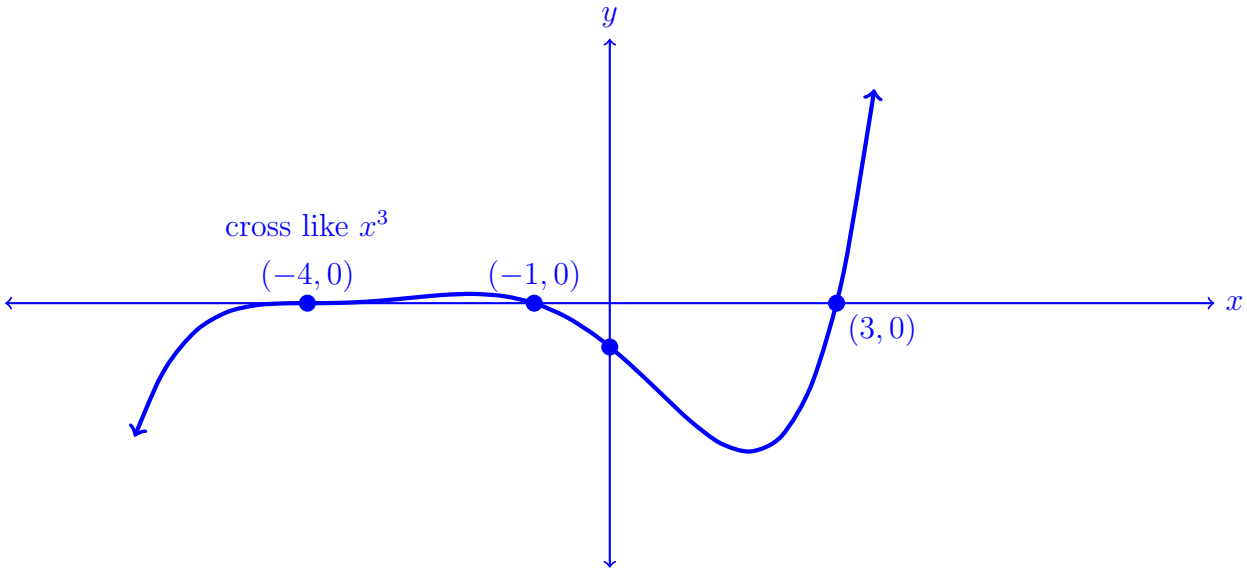
21. $y = (x-3)(x+1)(x+4)^3$

a) $(0,-1), (0,3)$ with multiplicity 1, and $(0,-4)$ with multiplicity 3

b) below

c) leading term: $(x)(x)(x)^3 = x^5$
positive, so RHS goes up
degree odd, so LHS is opposite (goes down)

d)



22. $y = (x-2)(x-1)^2(3x-7)$

a) $(0, 2), (0, 7/3)$ with multiplicity 1, and $(0, 1)$ with multiplicity 2

b) above

c) leading term: $(x)(x)^2(3x) = 3x^4$
positive, so RHS goes up
degree even, so LHS matches (goes up)

d)

